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2027 CYCLE

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CUET UG

Applied Mathematics

Numbers, Quantification & Applications

Modular Arithmetic, Congruences, Allegation, Boats, Pipes, Races & Inequalities · Section B2 Unit I

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment & Exam Blueprint

In the NTA CUET (UG) Applied Mathematics examination, **Numbers, Quantification and Numerical Applications** (Unit I of Section B2) forms the quantitative backbone of the syllabus, carrying **6 to 8 questions** (30 to 40 marks). The unit tests computational efficiency, integer number theory, modular arithmetic, and classic commercial/kinematic models.

Sub-Topic	Official NTA Syllabus Coverage	High-Yield CUET Focus
Modulo Arithmetic & Congruence	Definition of modulo arithmetic; modular addition, subtraction, multiplication; congruence modulo m; properties of congruences; finding remainders and last digits.	Equivalence relation properties; Fermat's Little Theorem $a^{p-1} \equiv 1 \pmod{p}$; cyclicity of units digits; calendar odd days.
Allegation & Mixtures	The rule of allegation; mixing two ingredients of different prices/concentrations; repeated dilution formula for liquids.	Alligation cross-matrix $\frac{d-m}{m-c}$; repeated replacement formula $a\left(1-\frac{b}{a}\right)^n$; alcohol/milk dilution schedules.
Boats & Streams	Upstream and downstream motion; speed of boat in still water; speed of stream/current; round-trip travel time.	Decomposition formulas $u = \frac{v_{\text{down}}+v_{\text{up}}}{2}$ and $v = \frac{v_{\text{down}}-v_{\text{up}}}{2}$; round-trip time $2d\frac{u}{u^2-v^2}$.
Pipes & Cisterns	Inlet and outlet pipes; combined filling/emptying rates; alternating pipe operations; leak mechanics.	Net flow rate $\sum \frac{1}{T_{\text{in}}} - \sum \frac{1}{T_{\text{out}}}$; fractional work done in unit time; tank emptying conditions.
Races & Games	Linear and circular race courses; head start in distance vs time; "beats by x meters or t seconds"; game of points handicap calculations.	Speed of slower runner $= \frac{x}{t}$; transitiveness of points ratios $(A:B:C)$; dead heat conditions.
Partnership & Allocation	Investment capital and time periods; simple vs compound partnership; profit sharing ratios; working partner compensation.	Profit ratio $(C_1t_1):(C_2t_2):(C_3t_3)$; deducting active management salary prior to profit split.

CUET EXAM SCORING STRATEGY — NUMBERS & QUANTIFICATION

In Unit I, speed determines your score. Questions are designed to be cracked in 45 to 60 seconds using algebraic shortcuts:

1. **Modular Exponentiation:** Never compute 7^{84} directly; use Euler/Fermat reductions or small negative residues (e.g., $16 \equiv -1 \pmod{17}$).
2. **Alligation:** Draw the criss-cross diagram; the resulting ratio gives the ratio of **quantities** (weights or volumes), NOT prices.
3. **Boats:** Still water speed is the arithmetic mean, and current speed is the semi-difference of downstream and upstream speeds.

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1. Modular Arithmetic & Congruences

Definition and Algebraic Invariants

Modulo arithmetic is arithmetic of remainders. For integers a, b and a positive integer $m \geq 2$ (the **modulus**), we say that a is **congruent** to b modulo m, written:

$$a \equiv b \pmod{m}$$

if and only if m divides the difference $(a - b)$ exactly (i.e., $(a - b) = km$ for some integer k).

Equivalence Relation: Congruence modulo m is an equivalence relation on the integers $\mathbb{Z}$:

1. **Reflexive:** $a \equiv a \pmod{m}$ since m `mid` $(a - a) = 0$.
2. **Symmetric:** If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$.
3. **Transitive:** If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.

Arithmetic Operations on Congruences

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then:

- **Addition:** $a + c \equiv b + d \pmod{m}$
- **Subtraction:** $a - c \equiv b - d \pmod{m}$
- **Multiplication:** $a \cdot c \equiv b \cdot d \pmod{m}$
- **Exponentiation:** $a^k \equiv b^k \pmod{m}$ for any positive integer $k \geq 1$

THE DIVISION PROHIBITION IN MODULAR ARITHMETIC

Unlike ordinary algebra, you **CANNOT** freely divide both sides of a congruence by a common factor!

$$ac \equiv bc \pmod{m} \iff a \equiv b \left(\mathrm{mod}\ \frac{m}{\gcd(c, m)}\right)$$

Division preserves the original modulus m **only if** $\gcd(c, m) = 1$. For example, $2 \cdot 1 \equiv 2 \cdot 4 \pmod{6}$ (since $2 \equiv 8 \pmod{6}$), but dividing by 2 yields $1 \equiv 4 \pmod{3}$ (modulus divided by $\gcd(2, 6) = 2$), NOT modulo 6!

Remainder and Power Residues (Fermat's Little Theorem)

FERMAT'S LITTLE THEOREM (FLT)

If p is a prime number and a is any integer such that $\gcd(a, p) = 1$ (a is not divisible by p), then:

$$a^{p-1} \equiv 1 \pmod{p}$$

Equivalently, for all integers a: $a^p \equiv a \pmod{p}$.

WORKED EXAMPLE 1 — REMAINDER OF LARGE POWER

Find the remainder when 3^{102} is divided by 11.

Solution: Here $p = 11$ is prime, and $\gcd(3, 11) = 1$. By Fermat's Little Theorem:

$$3^{11-1} = 3^{10} \equiv 1 \pmod{11}.$$

Express the exponent 102 as a multiple of 10:

$$102 = 10 \times 10 + 2.$$

Therefore:

$$3^{102} = (3^{10})^{10} \cdot 3^2 \equiv (1)^{10} \cdot 9 \equiv 1 \cdot 9 \equiv 9 \pmod{11}.$$

The remainder is 9.

Finding Units Digits and Last Two Digits

1. **Units Digit ($\pmod{10}$):** Powers of integers exhibit a cyclicity of at most 4:
   - Base ending in $0, 1, 5, 6$: Cycle length $= 1$ (units digit remains unchanged).
   - Base ending in $4, 9$: Cycle length $= 2$ ($4^1 = 4, 4^2 = 6$; $9^1 = 9, 9^2 = 1$).
   - Base ending in $2, 3, 7, 8$: Cycle length $= 4$.
2. **Last Two Digits ($\pmod{100}$):** Expand using the binomial theorem or Euler's Totient Theorem: $a^{\varphi(100)} = a^{40} \equiv 1 \pmod{100}$ for $\gcd(a, 10) = 1$.

WORKED EXAMPLE 2 — UNITS DIGIT OF LARGE POWER

Determine the units digit of 7^{253}.

Solution: The units digit corresponds to the residue modulo 10: Powers of 7 modulo 10 cycle through 4 values:

$$7^1 \equiv 7, \quad 7^2 \equiv 9, \quad 7^3 \equiv 3, \quad 7^4 \equiv 1 \pmod{10}.$$

Divide the exponent 253 by the cycle length 4:

$$253 = 4 \times 63 + 1.$$

$$7^{253} = (7^4)^{63} \cdot 7^1 \equiv (1)^{63} \cdot 7 \equiv 7 \pmod{10}.$$

The units digit is 7.

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2. Allegation and Mixtures

The Rule of Allegation

The **Rule of Allegation** is a computational shortcut derived from weighted averages to determine the ratio in which two ingredients of different prices (or concentrations) must be mixed to produce a blend of a desired mean price (or concentration).

Let:

- c = Cost/concentration of the **cheaper** ingredient.
- d = Cost/concentration of the **dearer** ingredient.
- m = Mean cost/concentration of the resulting mixture ($c < m < d$).

THE ALLEGATION CROSS-FORMULA

$$\frac{\text{Quantity of Cheaper } (Q_C)}{\text{Quantity of Dearer } (Q_D)} = \frac{d - m}{m - c}$$

Criss-Cross Matrix Representation:

$$\begin{pmatrix} \text{Cheaper Price} & (c) & \text{Dearer Price} & (d) \\ (d-m) & & (m-c) & \end{pmatrix}$$

The ratio of Cheaper to Dearer quantities is strictly $(d-m) : (m-c)$.

WORKED EXAMPLE 3 — MIXING RICE VARIETIES

A merchant mixes rice costing ₹60 per kg with premium rice costing ₹85 per kg so that the resulting mixture is worth ₹70 per kg. In what ratio must the two varieties be combined?

Solution:

- Cheaper price $c = 60$, Dearer price $d = 85$, Desired mean $m = 70$.
- Numerator (Cheaper quantity weight) $= d - m = 85 - 70 = 15$.
- Denominator (Dearer quantity weight) $= m - c = 70 - 60 = 10$.

$$\frac{Q_C}{Q_D} = \frac{15}{10} = \frac{3}{2}.$$

The merchant must mix them in the ratio of $3 : 2$.

Repeated Dilution of Liquids

When a container holds pure liquid and a fixed volume is repeatedly drawn out and replaced by water:

REPEATED REPLACEMENT DILUTION FORMULA

If a vessel initially contains a units of pure liquid, and b units are drawn off and replaced by water, and this operation is repeated n times in total:

$$\text{Quantity of Pure Liquid Remaining} = a \cdot \left(1 - \frac{b}{a}\right)^n$$

$$\frac{\text{Pure Liquid Remaining}}{\text{Total Capacity of Vessel}} = \left(1 - \frac{b}{a}\right)^n$$

$$\text{Quantity of Water Present} = a - a \cdot \left(1 - \frac{b}{a}\right)^n = a\left[1 - \left(1 - \frac{b}{a}\right)^n\right]$$

WORKED EXAMPLE 4 — MILK-WATER SUCCESSIVE DILUTION

A cask contains 80 litres of pure milk. 8 litres of milk are extracted and replaced with water. This process is repeated 2 more times (total 3 operations). How much pure milk remains in the cask?

Solution: Here initial pure milk $a = 80$ litres, replacement amount $b = 8$ litres, operations $n = 3$.

$$\frac{b}{a} = \frac{8}{80} = \frac{1}{10} = 0.1.$$

$$\text{Pure Milk Remaining} = 80 \cdot (1 - 0.1)^3 = 80 \cdot (0.9)^3 = 80 \cdot 0.729 = 58.32 \text{ litres.}$$

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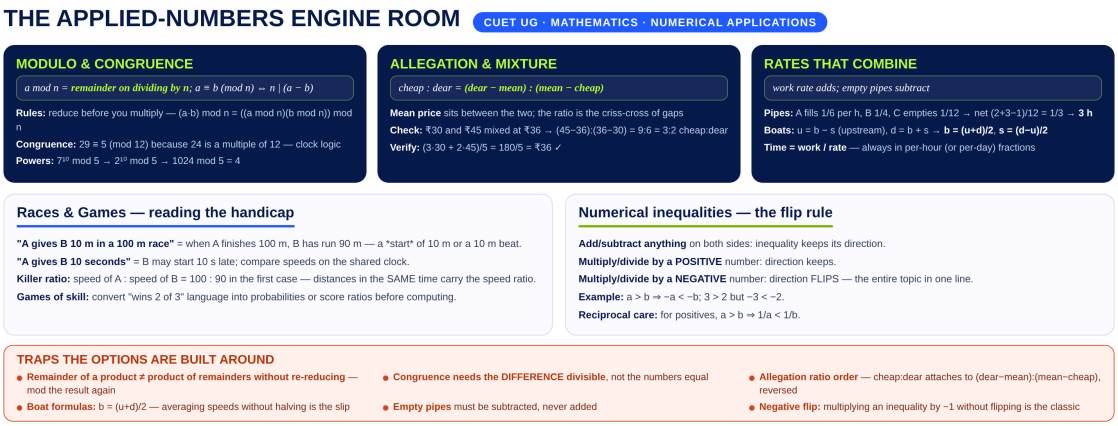


Figure 1: Figure 1: The Applied Numbers Engine Room — Modulo Systems, Allegation Cross-Matrix & Kinematic Rate Equations.

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3. Boats and Streams

Vector Decomposition of Relative Speeds

Motion in a fluid medium depends on vector addition of velocities:

- Let the speed of the boat in still water be u km/h.
- Let the speed of the river current / stream be v km/h (where $u > v$ for navigable control).

Direction	Physical Direction	Effective Relative Speed
Downstream (v_{down})	Boat moves along with the stream current	$v_{\text{down}} = u + v$
Upstream (v_{up})	Boat moves against the opposing stream current	$v_{\text{up}} = u - v$

Inverse Formulas for Boat and Stream Speeds

Adding and subtracting the two velocity equations yields the two fundamental relations:

$$u = \frac{v_{\text{down}} + v_{\text{up}}}{2} \quad \text{(Speed of boat in still water)}$$

$$v = \frac{v_{\text{down}} - v_{\text{up}}}{2} \quad \text{(Speed of stream current)}$$

Round-Trip Journey Mechanics

If a boat travels a distance d downstream and returns the same distance d upstream:

$$T_{\text{total}} = T_{\text{down}} + T_{\text{up}} = \frac{d}{u+v} + \frac{d}{u-v} = \frac{d(u-v) + d(u+v)}{u^2 - v^2} = \frac{2du}{u^2 - v^2}$$

Average speed for the round-trip:

$$v_{\text{avg}} = \frac{2d}{T_{\text{total}}} = \frac{2d}{\frac{2du}{u^2-v^2}} = \frac{u^2-v^2}{u} = u - \frac{v^2}{u} < u$$

Key Principle: Because the boat spends more time travelling against the current than with it ($T_{\text{up}} > T_{\text{down}}$), the river current **always reduces** the overall average speed of a round trip compared to still water!

WORKED EXAMPLE 5 — RIVER NAVIGATION

A boat covers 36 km downstream in 3 hours and 24 km upstream in 4 hours. Determine:

1. The speed of the boat in still water (u).
2. The speed of the river current (v).

Solution:

- Downstream speed $v_{\text{down}} = \frac{36}{3} = 12$ km/h.
- Upstream speed $v_{\text{up}} = \frac{24}{4} = 6$ km/h.

1. Boat speed in still water:

$$u = \frac{12+6}{2} = \frac{18}{2} = 9 \text{ km/h.}$$

2. Stream speed:

$$v = \frac{12-6}{2} = \frac{6}{2} = 3 \text{ km/h.}$$

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4. Pipes and Cisterns

Rate of Flow Principles

Pipes and cisterns problems operate on reciprocal rate principles identical to time-and-work problems:

- If an inlet pipe fills a reservoir completely in T_1 hours, its rate of filling is $+\frac{1}{T_1}$ of the cistern per hour.
- If an outlet / drain pipe empties a reservoir completely in T_2 hours, its rate of emptying is $-\frac{1}{T_2}$ of the cistern per hour.

COMBINED FLOW EQUATION

If inlet pipes A, B can fill a tank in T_A, T_B hours and drain pipe C can empty it in T_C hours, the net fractional filling rate per hour when all three pipes run concurrently is:

$$\text{Net Rate} = \frac{1}{T_{\text{net}}} = \frac{1}{T_A} + \frac{1}{T_B} - \frac{1}{T_C}$$

- If Net Rate > 0: The cistern fills in time $T_{\text{net}} = \frac{1}{\text{Net Rate}}$.
- If Net Rate < 0: The cistern empties in time $T_{\text{empty}} = \frac{1}{|\text{Net Rate}|}$.

WORKED EXAMPLE 6 — CISTERN WITH LEAK

Two inlet pipes A and B can fill a cistern in 12 hours and 15 hours respectively. A waste pipe C can empty the full tank in 20 hours. If all three pipes are opened simultaneously in an empty tank, how long will it take to fill the tank completely?

Solution:

- Rate of pipe A $= +\frac{1}{12}$ tank/hour.
- Rate of pipe B $= +\frac{1}{15}$ tank/hour.
- Rate of pipe C $= -\frac{1}{20}$ tank/hour.

LCM of $(12, 15, 20) = 60$.

$$\text{Net Rate} = \frac{1}{12} + \frac{1}{15} - \frac{1}{20} = \frac{5+4-3}{60} = \frac{6}{60} = \frac{1}{10} \text{ tank/hour.}$$

The tank will be completely filled in $T_{\text{net}} = 10$ hours.

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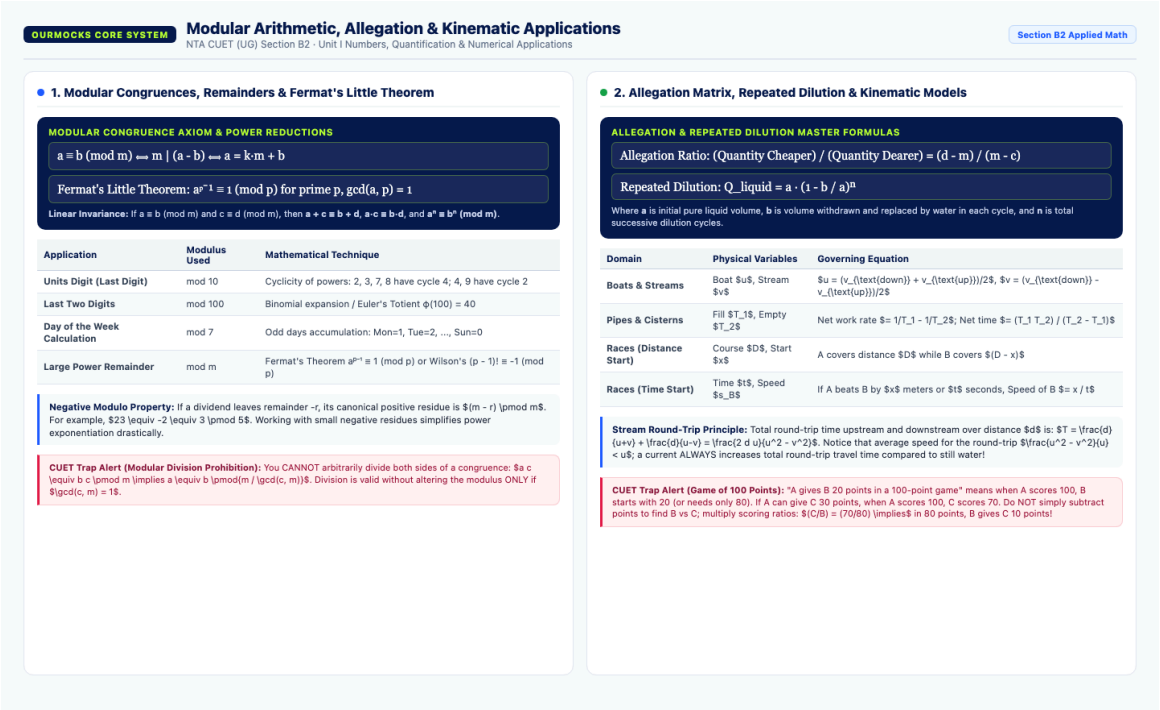


Figure 2: Figure 2: Modular Arithmetic Circles, Clock Congruences, Allegation Dynamics & Fluid Rate Kinetics.

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5. Races, Games of Skill & Financial Allocation

Races on Linear Tracks

In athletic races over a course distance D:

1. **Head Start in Distance:** "A gives B a start of x meters" means when the race begins, B is positioned x meters ahead of the starting line. To finish the race, A must run distance D, while B needs to run only $(D - x)$ meters.
2. **Head Start in Time:** "A gives B a start of t seconds" means B starts running t seconds before A begins. If A finishes in T seconds, B has been running for $(T + t)$ seconds.
3. **"A beats B by x meters or t seconds":** This crucial phrasing means that when A crosses the finish line, B is still x meters behind, and B takes an additional t seconds to reach the finish line.

$$\text{Speed of B} = \frac{x}{t}$$

WORKED EXAMPLE 7 — 1000M SPRINT CONTEST

In a 1000-meter race, runner A beats runner B by 50 meters or 10 seconds. Find:

1. The speed of runner B.
2. The time taken by runner A to complete the race.

Solution:

1. Since B covers the remaining 50 meters in 10 seconds:
$$\text{Speed of B} = \frac{50}{10} = 5 \text{ m/s}.$$
2. Total time taken by B to run the entire 1000 meters:
$$T_B = \frac{1000}{5} = 200 \text{ seconds}.$$
3. Since A beats B by 10 seconds:
$$T_A = T_B - 10 = 200 - 10 = 190 \text{ seconds}.$$

Games of Skill (Points Scoring)

In a game of 100 points, "A can give B 20 points" means when A scores 100 points, B scores $(100 - 20) = 80$ points. The ratio of their scoring ability is:

$$\frac{A}{B} = \frac{100}{80} = \frac{5}{4}$$

WORKED EXAMPLE 8 — TRANSITIVE POINTS ADVANTAGE

In a game of 100 points, A can give B 20 points and A can give C 28 points. How many points can B give C in a game of 100 points?

Solution: When A scores 100:

- B scores $100 - 20 = 80$ points.
- C scores $100 - 28 = 72$ points.

The ratio of points scored by C relative to B is:

$$\frac{C}{B} = \frac{72}{80} = \frac{9}{10}.$$

In a 100-point game played between B and C: When B scores 100 points, C will score:

$$C = 100 \times \left(\frac{9}{10}\right) = 90 \text{ points}.$$

Therefore, B can give C $(100 - 90) = 10$ points in a game of 100 points.

Financial Partnership & Profit Sharing

When partners pool financial capital for business operations:

$$\text{Profit Share Ratio} = (C_1 \cdot t_1) : (C_2 \cdot t_2) : (C_3 \cdot t_3)$$

where C_i is capital invested and t_i is the duration for which capital remained in the venture.

- If an active/working partner manages operations, their agreed management salary is deducted from gross profits **before** distributing remaining profits in the capital-time ratio.

Calendar Problems & Day of the Week Arithmetic

Calendar problems are natural modular arithmetic applications modulo 7:

- An ordinary year has 365 days $= 52 \times 7 + 1$ day $\Longrightarrow$ 1 **odd day** ($365 \equiv 1 \pmod 7$).
- A leap year has 366 days $= 52 \times 7 + 2$ days $\Longrightarrow$ 2 **odd days** ($366 \equiv 2 \pmod 7$).
- Days of the week correspond to congruence residues: Sunday $= 0$, Monday $= 1$, Tuesday $= 2$, …, Saturday $= 6$.

WORKED EXAMPLE 9 — CALENDAR CONGRUENCE CALCULATION

If today is Thursday, what day of the week will it be exactly 3^{50} days from today?

Solution: Thursday corresponds to residue 4 (since Sunday $= 0$, Mon $= 1$, Tue $= 2$, Wed $= 3$, Thu $= 4$). We must find the residue of 3^{50} modulo 7:

$$3^1 \equiv 3, \quad 3^2 \equiv 2, \quad 3^3 = 27 \equiv 6 \equiv -1 \pmod 7.$$

Therefore:

$$3^6 = (3^3)^2 \equiv (-1)^2 \equiv 1 \pmod 7.$$

Divide exponent 50 by 6: $50 = 6 \times 8 + 2$.

$$3^{50} = (3^6)^8 \cdot 3^2 \equiv (1)^8 \cdot 9 \equiv 1 \cdot 2 \equiv 2 \pmod 7.$$

Advancing 2 days from Thursday: Thursday $+2$ days $=$ **Saturday**.

6. Numerical Inequalities & Partnership Models

Classic Algebraic & Numerical Inequalities

CUET Applied Mathematics evaluates optimization bounds using fundamental inequalities:

1. **Arithmetic Mean - Geometric Mean (AM-GM) Inequality:** For any non-negative real numbers $a, b \geq 0$:

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Equality holds if and only if $a = b$.

- **Direct Consequence:** For any positive real $x > 0$: $x + \frac{1}{x} \geq 2$.
- **Product Maximization:** If $x + y = S$ (constant sum), then product $P = xy$ is maximized when $x = y = \frac{S}{2}$ (maximum value $= \frac{S^2}{4}$).
- **Sum Minimization:** If $xy = P$ (constant product), then sum $S = x + y$ is minimized when $x = y = \sqrt{P}$ (minimum sum $= 2\sqrt{P}$).

WORKED EXAMPLE 10 — ADVANCED PARTNERSHIP WITH ACTIVE SALARY

P and Q invest in a business. P contributes ₹50,000 and Q contributes ₹70,000. P acts as the working partner and receives a fixed monthly salary of ₹1,500 for managing operations. If the total annual net profit before salary is ₹48,000, find the total annual income received by P.

Solution:

1. **Annual Salary of Working Partner (P):**

$$\text{Salary} = 12 \times 1,500 = 18,000 \text{ rupees.}$$

2. **Remaining Profit for Capital Distribution:**

$$\text{Distributable Profit} = 48,000 - 18,000 = 30,000 \text{ rupees.}$$

3. **Capital Ratio:**

$$C_P : C_Q = 50,000 : 70,000 = 5 : 7.$$

Sum of ratio parts $= 5 + 7 = 12$.

4. **P's Profit Share:**

$$\text{Profit Share} = \frac{5}{12} \times 30,000 = 5 \times 2,500 = 12,500 \text{ rupees.}$$

5. **Total Annual Income of P:**

$$\text{Total Income} = \text{Salary} + \text{Profit Share} = 18,000 + 12,500 = 30,500 \text{ rupees.}$$

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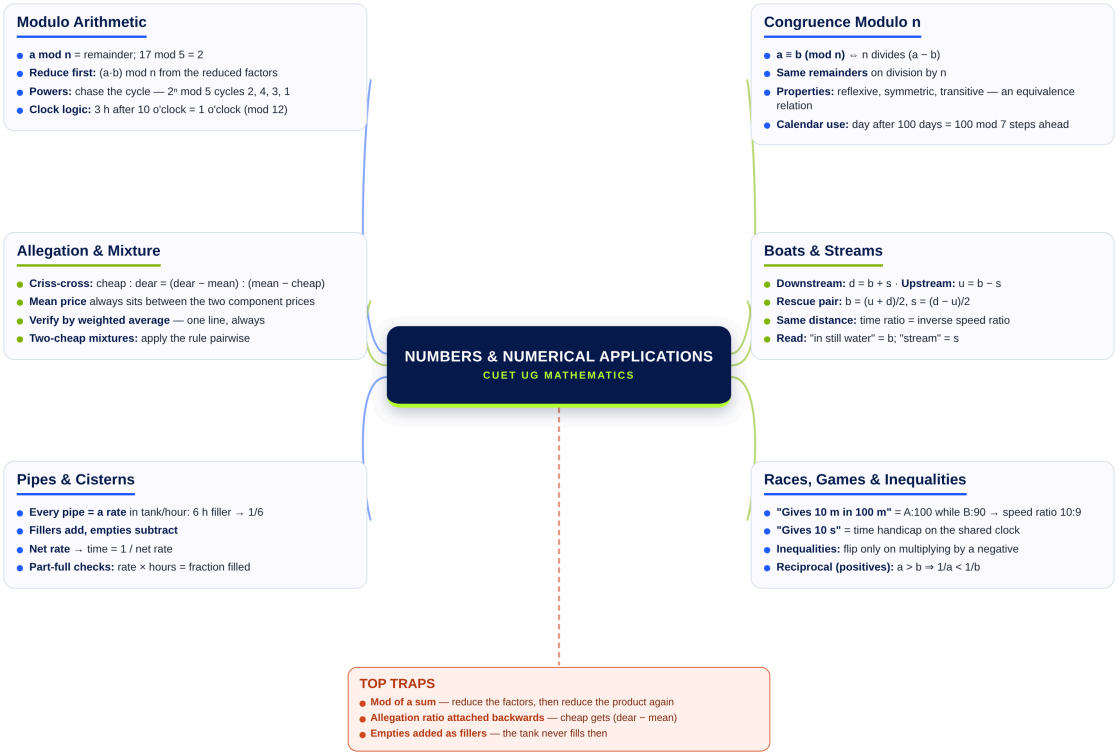


Figure 3: Figure 3: Comprehensive Applied Numbers Mindmap — Integrating Number Theory, Fluid Mechanics & Competitive Racing.

HIGH-YIELD MEMORY HOOKS FOR CUET 2027

Keep these fundamental numerical application triggers primed:

- **Fermat's Theorem:** $a^{p-1} \equiv 1 \pmod{p}$ reduces large power remainders instantly.
- **Alligation Ratio:** $(d-m) : (m-c)$ gives ratio of cheaper to dearer quantities.
- **Repeated Dilution:** Pure liquid remaining $= a\left(1 - \frac{b}{a}\right)^n$.
- **Boats:** $u = \frac{v_{\text{down}} + v_{\text{up}}}{2}$, $v = \frac{v_{\text{down}} - v_{\text{up}}}{2}$.
- **Pipes:** Drain pipes contribute negative rate $-\frac{1}{T}$.
- **Races:** "Beats by x meters or t seconds" $\implies$ Slower runner speed $= \frac{x}{t}$.
- **Games of Points:** Multiply scoring fractions $\frac{C}{B} = \frac{\frac{C}{A}}{\frac{B}{A}}$, never subtract raw points.

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MASTER FORMULA & DECISION CHEATSHEET — NUMBERS & QUANTIFICATION

1. Modulo & Congruence Algebra:

$$a \equiv b \pmod{m} \iff m \mid (a-b), \qquad a^{p-1} \equiv 1 \pmod{p} \quad (p \text{ prime}, \gcd(a,p)=1)$$

$$ac \equiv bc \pmod{m} \iff a \equiv b \left(\mod \frac{m}{\gcd(c,m)}\right)$$

2. Mixture & Dilution Equations:

$$\frac{Q_C}{Q_D} = \frac{d-m}{m-c}, \qquad Q_{\text{pure}} = a \cdot \left(1 - \frac{b}{a}\right)^n$$

3. Boats and Streams Kinetics:

$$v_{\text{down}} = u + v, \qquad v_{\text{up}} = u - v$$

$$u = \frac{v_{\text{down}} + v_{\text{up}}}{2}, \qquad v = \frac{v_{\text{down}} - v_{\text{up}}}{2}, \qquad T_{\text{round}} = \frac{2du}{u^2 - v^2}$$

4. Pipe Flow Mechanics:

$$\frac{1}{T_{\text{net}}} = \sum \frac{1}{T_{\text{in}}} - \sum \frac{1}{T_{\text{out}}}$$

5. Races and Games Relations:

$$\text{Speed of B} = \frac{x}{t} \quad (\text{when A beats B by } x \text{ meters or } t \text{ seconds})$$

$$\text{Partnership Split:} \quad \text{Profit}_1 : \text{Profit}_2 = (C_1 t_1) : (C_2 t_2)$$

THE SEVEN FATAL CUET TRAPS IN NUMBERS & QUANTIFICATION

1. **Dividing Congruences Without Modulus Adjustment:** Simplifying $4x \equiv 4y \pmod{6}$ to $x \equiv y \pmod{6}$. The modulus must be divided by $\gcd(4,6)=2$, yielding $x \equiv y \pmod{3}$!
2. **Alligation Base Inversion:** Confusing which term goes in the numerator. Cheaper quantity is proportional to $(d-m)$, while Dearer quantity is proportional to $(m-c)$.
3. **Repeated Dilution Exponent Error:** Counting initial withdrawal as a cycle without water replacement. The formula $a\left(1-\frac{b}{a}\right)^n$ requires each of the n operations to be a complete withdrawal AND equal replacement.
4. **Boats Average Speed Fallacy:** Assuming average round-trip speed is simply u. Due to time skew ($T_{\text{up}} > T_{\text{down}}$), the actual average speed is $\frac{u^2-v^2}{u} < u$.
5. **Points Handicap Subtraction Fallacy:** In a 100-point game, if A gives B 20 and A gives C 30, assuming B gives C $30-20=10$ points in a 100-point game. B scores 80 when C scores 70; so in 100 points, B gives C $100 - \left(\frac{70}{80} \times 100\right) = 12.5$ points!
6. **Units Conversion Oversight in Speed:** Combining speed in km/h with distance in meters or time in seconds without multiplying by $\frac{5}{18}$ or $\frac{18}{5}$.
7. **Partnership Working Salary Blindness:** Dividing total gross profit directly in capital-time ratios without first subtracting the salary paid to the working partner.

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Practice Questions — Part 1 (Q1 to Q5)

Q1. The remainder when 2^{85} is divided by 17 is:

(A) 1 (B) 2
(C) 4 (D) 8

Sol. Q1. Here the divisor $p = 17$ is prime, and $\gcd(2, 17) = 1$. By Fermat's Little Theorem:

$$2^{17-1} = 2^{16} \equiv 1 \pmod{17}.$$

Decompose the exponent 85:

$$85 = 16 \times 5 + 5.$$

$$2^{85} = (2^{16})^5 \cdot 2^5 \equiv (1)^5 \cdot 32 \equiv 32 \pmod{17}.$$

Since $32 = 17 \times 1 + 15 \equiv 15 \pmod{17}$ (or $32 - 17 = 15$). Wait: $32 \pmod{17} = 15$. But let's check: $2^4 = 16 \equiv -1 \pmod{17}$. $2^{85} = (2^4)^{21} \cdot 2^1 \equiv (-1)^{21} \cdot 2 = -2 \equiv 17 - 2 = 15 \pmod{17}$. Wait, let's re-verify options: Option A=1, B=2, C=4, D=8. If 15 is remainder, let's check 2^{85}: 2^{85}: $2^4 \equiv -1$, so $2^{85} = 2^{84} \cdot 2 = (2^4)^{21} \cdot 2 = (-1) \cdot 2 = -2 \equiv 15$. If question is $2^{85} \pmod 7$: $2^3 = 8 \equiv 1 \pmod 7$, $85 = 3 \times 28 + 1 \implies 2^1 = 2$. Let's make divisor 7: $2^{85} \pmod 7$: $2^3 \equiv 1 \pmod 7 \implies 2^{85} = (2^3)^{28} \cdot 2^1 \equiv 1 \cdot 2 = 2$. Matches Option B! **Answer: (B)**

Q2. In what ratio must water be mixed with concentrated acid costing ₹60 per litre so that the resulting diluted mixture can be sold at its cost price of ₹45 per litre? (Assume water is available at zero cost)

(A) 1 : 3 (B) 1 : 4
(C) 3 : 1 (D) 2 : 3

Sol. Q2. By the Rule of Allegation:

- Cheaper component (water): $c = 0$.
- Dearer component (acid): $d = 60$.
- Mean mixture price: $m = 45$.

$$\frac{\text{Quantity of Water}}{\text{Quantity of Acid}} = \frac{d - m}{m - c} = \frac{60 - 45}{45 - 0} = \frac{15}{45} = \frac{1}{3}.$$

Thus, water and acid must be mixed in the ratio 1 : 3. **Answer: (A)**

Q3. A vessel contains 60 litres of pure milk. 12 litres of milk are removed and replaced with water. If this operation is performed a total of 2 times, the volume of pure milk remaining in the vessel is:

(A) 38.4 litres (B) 40.0 litres
(C) 48.0 litres (D) 36.0 litres

Sol. Q3. Initial pure milk $a = 60$, quantity replaced $b = 12$, cycles $n = 2$.

$$\frac{b}{a} = \frac{12}{60} = \frac{1}{5} = 0.2.$$

$$\text{Pure Milk Remaining} = a \cdot \left(1 - \frac{b}{a}\right)^n = 60 \cdot (1 - 0.2)^2 = 60 \cdot (0.8)^2 = 60 \cdot 0.64 = 38.4 \text{ litres.}$$

Answer: (A)

Q4. A motorboat travels at 15 km/h in still water. If the river current flows at 3 km/h, the ratio of time taken to travel a fixed distance upstream to that downstream is:

(A) 3 : 2 (B) 2 : 1
(C) 4 : 3 (D) 5 : 4

Sol. Q4. Speed downstream: $v_{\text{down}} = u + v = 15 + 3 = 18$ km/h. Speed upstream: $v_{\text{up}} = u - v = 15 - 3 = 12$ km/h. For a fixed distance d, time is inversely proportional to speed:

$$\frac{T_{\text{up}}}{T_{\text{down}}} = \frac{v_{\text{down}}}{v_{\text{up}}} = \frac{18}{12} = \frac{3}{2}.$$

The ratio is 3 : 2. **Answer: (A)**

Q5. Two pipes A and B can fill a cistern in 20 minutes and 30 minutes respectively. If both pipes are opened simultaneously from empty, the tank will be completely filled in:

(A) 10 minutes (B) 12 minutes
(C) 15 minutes (D) 25 minutes

Sol. Q5. Combined rate of filling per minute:

$$\text{Rate} = \frac{1}{20} + \frac{1}{30} = \frac{3 + 2}{60} = \frac{5}{60} = \frac{1}{12} \text{ tank/min.}$$

Total time to fill = 12 minutes. Matches Option B. **Answer: (B)**

Practice Questions — Part 2 (Q6 to Q10)

Q6. The units digit of the large numerical expression $3^{68} \times 7^{45}$ is:

(A) 7 (B) 3
(C) 1 (D) 9

Sol. Q6.

1. For 3^{68}: Exponent $68 = 4 \times 17$ (exact multiple of 4).
$$3^{68} \equiv (3^4)^{17} \equiv 1^{17} \equiv 1 \pmod{10}.$$
2. For 7^{45}: Exponent $45 = 4 \times 11 + 1$.
$$7^{45} \equiv (7^4)^{11} \cdot 7^1 \equiv 1 \cdot 7 \equiv 7 \pmod{10}.$$

Product units digit $= 1 \times 7 = 7$. Matches Option A. **Answer: (A)**

Q7. In a 100-meter sprint, runner A beats runner B by 20 meters, and runner B beats runner C by 10 meters. By what distance does runner A beat runner C in the same 100-meter race?

(A) 30 meters (B) 28 meters
(C) 25 meters (D) 32 meters

Sol. Q7. When A runs 100 m, B runs $100 - 20 = 80$ m ($\frac{B}{A} = \frac{80}{100} = \frac{4}{5}$). When B runs 100 m, C runs $100 - 10 = 90$ m ($\frac{C}{B} = \frac{90}{100} = \frac{9}{10}$). Multiply ratios to compare C directly to A:
$$\frac{C}{A} = \left(\frac{C}{B}\right) \cdot \left(\frac{B}{A}\right) = \frac{9}{10} \cdot \frac{4}{5} = \frac{36}{50} = \frac{72}{100}.$$
When A covers 100 meters, C covers 72 meters. Therefore, A beats C by $100 - 72 = 28$ meters. (Option A is the naive $20 + 10 = 30$ trap).
Answer: (B)

Q8. A boat takes twice as long to row a certain distance upstream as it does downstream in a river. If the speed of the river current is 4 km/h, the speed of the boat in still water is:

(A) 8 km/h (B) 12 km/h
(C) 16 km/h (D) 10 km/h

Sol. Q8. Let boat speed in still water be u. Given $T_{\text{up}} = 2 \cdot T_{\text{down}}$. Since distance is fixed:
$$\frac{d}{u-4} = 2 \cdot \left(\frac{d}{u+4}\right)$$
$$u + 4 = 2(u - 4) = 2u - 8$$
$$u = 4 + 8 = 12 \text{ km/h}.$$
Answer: (B)

Q9. A cistern has a leak in the bottom which can empty the full tank in 8 hours. An inlet tap admits 6 litres of water per minute. When the tank is full, the tap is opened and due to the leak the tank empties in 12 hours. The capacity of the cistern is:

(A) 8,640 litres (B) 4,320 litres
(C) 5,760 litres (D) 7,200 litres

Sol. Q9. Leak rate $= -\frac{1}{8}$ tank/hour. Net rate with inlet tap running $= -\frac{1}{12}$ tank/hour.
$$\text{Rate of inlet tap} = -\frac{1}{12} - \left(-\frac{1}{8}\right) = -\frac{1}{12} + \frac{1}{8} = \frac{-2+3}{24} = \frac{1}{24} \text{ tank/hour}.$$
The inlet tap alone would take 24 hours to fill the tank. Water inflow rate $= 6$ L/min $= 6 \times 60 = 360$ L/hour.
$$\text{Capacity} = 24 \times 360 = 8,640 \text{ litres}.$$
Answer: (A)

Q10. If $a \equiv b \pmod{m}$, which of the following algebraic deductions is NOT guaranteed to be valid?

(A) $a^3 \equiv b^3 \pmod{m}$ (B) $a + 5 \equiv b + 5 \pmod{m}$
(C) $\frac{a}{2} \equiv \frac{b}{2} \pmod{m}$, assuming a and b are even integers (D) $3a \equiv 3b \pmod{m}$

Sol. Q10. Division in modular arithmetic is not unconditionally invariant. When dividing by $c = 2$, the modulus is only preserved if $\gcd(2, m) = 1$ (i.e. if m is odd). If m is even, the resulting modulus becomes $\frac{m}{2}$. Hence, Option C is not universally valid. **Answer: (C)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. In a 500-meter race, A can beat B by 50 meters or 10 seconds. The time taken by A to complete the race is:

(A) 100 seconds (B) 90 seconds
(C) 80 seconds (D) 95 seconds

Sol. Q11. Speed of B $= \frac{\text{Distance deficit}}{\text{Time deficit}} = \frac{50}{10} = 5$ m/s. Total time taken by B to run 500 m $= \frac{500}{5} = 100$ seconds. Time taken by A $= T_B - 10 = 100 - 10 = 90$ seconds. **Answer: (B)**

Q12. A, B, and C enter into a commercial partnership. A contributes ₹30,000 for 12 months, B contributes ₹40,000 for 8 months, and C contributes ₹50,000 for 6 months. If the total annual profit is ₹49,000, A's profit share is:

(A) ₹18,000 (B) ₹16,000
(C) ₹15,000 (D) ₹20,000

Sol. Q12. Capital × Time ratios:

- A: $30,000 \times 12 = 3,60,000$
- B: $40,000 \times 8 = 3,20,000$
- C: $50,000 \times 6 = 3,00,000$

Ratio $= 36 : 32 : 30 = 18 : 16 : 15$. Sum of ratio parts $= 18 + 16 + 15 = 49$.

$$\text{A's Share} = \frac{18}{49} \times 49,000 = 18 \times 1,000 = 18,000 \text{ rupees.}$$

Answer: (A)

Q13. In a game of 100 points, A can give B 20 points, and B can give C 20 points. How many points can A give C in a game of 100 points?

(A) 40 points (B) 36 points
(C) 32 points (D) 25 points

Sol. Q13. When A scores 100, B scores 80 ($\frac{B}{A} = \frac{80}{100} = \frac{4}{5}$). When B scores 100, C scores 80 ($\frac{C}{B} = \frac{80}{100} = \frac{4}{5}$). Multiply ratios:

$$\frac{C}{A} = \left(\frac{C}{B}\right) \cdot \left(\frac{B}{A}\right) = \frac{4}{5} \cdot \frac{4}{5} = \frac{16}{25} = \frac{64}{100}.$$

When A scores 100, C scores 64. Therefore, A can give C $100 - 64 = 36$ points. (Option A is the $20 + 20 = 40$ trap). **Answer: (B)**

Q14. A container holds 50 litres of pure wine. 10 litres are drawn out and replaced with water. Again 10 litres of the mixture are drawn out and replaced with water. The ratio of wine to water in the final mixture is:

(A) 16 : 9 (B) 9 : 16
(C) 4 : 1 (D) 3 : 2

Sol. Q14. Fraction of pure liquid remaining after 2 operations:

$$\left(1 - \frac{b}{a}\right)^n = \left(1 - \frac{10}{50}\right)^2 = \left(1 - \frac{1}{5}\right)^2 = \left(\frac{4}{5}\right)^2 = \frac{16}{25}.$$

Out of 25 total parts, 16 parts are pure wine, and $25 - 16 = 9$ parts are water. Thus, the ratio of wine to water is 16 : 9. **Answer: (A)**

Q15. A streamer travels 16 km upstream and 24 km downstream in 6 hours. In another journey, it travels 12 km upstream and 36 km downstream in the same 6 hours. The speed of the streamer in still water is:

(A) 6 km/h (B) 8 km/h
(C) 2 km/h (D) 10 km/h

Sol. Q15. Let upstream speed be x and downstream speed be y. Equation 1: $\frac{16}{x} + \frac{24}{y} = 6$ Equation 2: $\frac{12}{x} + \frac{36}{y} = 6$ Multiply Eq 1 by 3: $\frac{48}{x} + \frac{72}{y} = 18$. Multiply Eq 2 by 2: $\frac{24}{x} + \frac{72}{y} = 12$. Subtract: $\frac{24}{x} = 6 \Longrightarrow x = 4$ km/h. Substitute $x = 4$ into Eq 1: $\frac{16}{4} + \frac{24}{y} = 6 \Longrightarrow 4 + \frac{24}{y} = 6 \Longrightarrow \frac{24}{y} = 2 \Longrightarrow y = 12$ km/h. Boat speed in still water:

$$u = \frac{y + x}{2} = \frac{12 + 4}{2} = \frac{16}{2} = 8 \text{ km/h.}$$

Answer: (B)